

Program4:

Integrate git with Docker for source code controlled application

Project structure

- |
- |--Dockerfile
- |--[index.js](#)
- |--package.json
- |--package-lock.json

Step1: Create a folder program4

Step2: create a package.json command using npm init -y

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab$ mkdir program4
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab$ cd program4
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$ npm init -y
Wrote to /home/1rv24mc055_kishan_s/lab/program4/package.json:
```

```
{
  "name": "program4",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC"
}
```

Step3: Create a [index.js](#) file

```
GNU nano 7.2                                index.js *
const express=require("express");
const app=express();
const PORT=3000;
app.get("/",(req,res)=>{
    res.send("program 4");
}):

app.listen(PORT,()=>{
console.log(`on port ${PORT}`);
```

Step4 :Add the express dependency

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$ npm i express
```

added 65 packages, and audited 66 packages in 4s

22 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

Step5: Create a Dockerfile

```
#stage 1 build stage

FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

#stage 2 Runtime stage

FROM node:18-slim
WORKDIR /app
copy --from=build /app /app
EXPOSE 3000
CMD ["node","index.js"]
```

Step6 : build the image

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$ docker build -t program4 .
[+] Building 44.6s (15/15) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 264B                                0.0s
=> WARN: ConsistentInstructionCasing: Command 'copy' should match the case of the comm 0.0s
=> [internal] load metadata for docker.io/library/node:18-slim    4.0s
=> [internal] load metadata for docker.io/library/node:18         0.0s
=> [auth] library/node:pull token for registry-1.docker.io        0.0s
=> [internal] load .dockerignore                                    0.0s
=> => transferring context: 2B                                       0.0s
=> [build 1/5] FROM docker.io/library/node:18                      0.1s
=> [internal] load build context                                    0.2s
=> => transferring context: 2.14MB                                    0.2s
=> [stage-1 1/3] FROM docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b29 40.0s
=> => resolve docker.io/library/node:18-slim@sha256:f9ab18e354e6855ae56ef2b290dd225c1e 0.0s
=> => sha256:b98d3ae1ab80d768fc7be41c07c4757d205cc33d2d26261b8273defb4 3.31kB / 3.31kB 1.2s
=> => sha256:b1831021e35a69864dfc4c89eab9ab1232cf3508c787b082c55ee3 38.25MB / 38.25MB 38.1s
=> => sha256:f9ab18e354e6855ae56ef2b290dd225c1e51a564f87584b9bd21dd651 6.49kB / 6.49kB 0.0s
=> => sha256:fc3faf127a182135fd956e68d570b1932a758f8008866d8dd6e131cf8 1.93kB / 1.93kB 0.0s
=> => sha256:101e0128c8ea90af6e5eba2abbae8486503c6383c35cb30e2c60842a5 6.54kB / 6.54kB 0.0s
=> => sha256:61320b01ae5e0798393ef25f2dc72faf43703e60ba089b07d7170a 28.23MB / 28.23MB 18.9s
=> => sha256:c768ab8cba73bb84835e67ed974c93e3e0e3ce1f67a73802422d66021 1.71MB / 1.71MB 3.7s
=> => sha256:8c994cf49dd19aec88c96926e1a13b36a8f3e159942a5a2d0a9e2dc3e9e6c 448B / 448B 4.5s
=> => extracting sha256:61320b01ae5e0798393ef25f2dc72faf43703e60ba089b07d7170acbabbf8f 1.3s
=> => extracting sha256:b98d3ae1ab80d768fc7be41c07c4757d205cc33d2d26261b8273defb454553 0.0s
=> => extracting sha256:b1831021e35a69864dfc4c89eab9ab1232cf3508c787b082c55ee386c5c852 1.6s
=> => extracting sha256:c768ab8cba73bb84835e67ed974c93e3e0e3ce1f67a73802422d66021fc9a0 0.1s
=> => extracting sha256:8c994cf49dd19aec88c96926e1a13b36a8f3e159942a5a2d0a9e2dc3e9e6c3 0.0s
=> [build 2/5] WORKDIR /app                                         0.1s
=> [build 3/5] COPY package*.json ./                                0.0s
=> [build 4/5] RUN npm install                                      19.1s
=> [build 5/5] COPY . .                                             0.4s
=> [stage-1 2/3] WORKDIR /app                                       0.2s
=> [stage-1 3/3] COPY --from=build /app /app                        0.1s
=> exporting to image                                               0.1s
=> => exporting layers                                              0.1s
=> => writing image sha256:f83841254baf349806cfaa1a6971de5bcf2311409f7d5ca706e6203e5c9 0.0s
=> => naming to docker.io/library/program4                          0.0s
```

Step7: Login to github and create access token and create a new repository

Step8: Initialize the git repository locally

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$ git add .
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$ git commit -m "Initial commit"
[master (root-commit) f620634] Initial commit
587 files changed, 61342 insertions(+)
create mode 100644 Dockerfile
create mode 100644 index.js
create mode 100644 node_modules/.package-lock.json
create mode 100644 node_modules/accepts/HISTORY.md
create mode 100644 node_modules/accepts/LICENSE
create mode 100644 node_modules/accepts/README.md
create mode 100644 node_modules/accepts/index.js
create mode 100644 node_modules/accepts/package.json
create mode 100644 node_modules/body-parser/LICENSE
create mode 100644 node_modules/body-parser/README.md
create mode 100644 node_modules/body-parser/index.js
create mode 100644 node_modules/body-parser/lib/read.js
create mode 100644 node_modules/body-parser/lib/types/json.js
create mode 100644 node_modules/body-parser/lib/types/raw.js
create mode 100644 node_modules/body-parser/lib/types/text.js
create mode 100644 node_modules/body-parser/lib/types/urlencoded.js
create mode 100644 node_modules/body-parser/lib/utils.js
```

Step9 :Connect to remote repository

```
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$ git branch -M main
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$ git remote add origin https://github.com
/Kishankharvi/program-04.git
1rv24mc055_kishan_s@kishan-L0Q-15IAX9:~/lab/program4$ git push -u origin main
Username for 'https://github.com': kishankharvi
Password for 'https://kishankharvi@github.com':
Enumerating objects: 676, done.
Counting objects: 100% (676/676), done.
Delta compression using up to 12 threads
Compressing objects: 100% (624/624), done.
Writing objects: 100% (676/676), 664.70 KiB | 3.71 MiB/s, done.
Total 676 (delta 118), reused 0 (delta 0), pack-reused 0
remote: Resolving deltas: 100% (118/118), done.
To https://github.com/Kishankharvi/program-04.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
```

Output

The screenshot shows the GitHub interface for a repository named 'program-04' by user 'Kishankharvi'. The repository is public and has 0 stars, 0 forks, and 0 watchers. The main branch is selected. The file list shows the initial commit with the following files: node_modules, Dockerfile, index.js, package-lock.json, and package.json. The 'About' section on the right indicates that no description, website, or topics are provided.

File	Commit	Time
node_modules	Initial commit	9 minutes ago
Dockerfile	Initial commit	9 minutes ago
index.js	Initial commit	9 minutes ago
package-lock.json	Initial commit	9 minutes ago
package.json	Initial commit	9 minutes ago